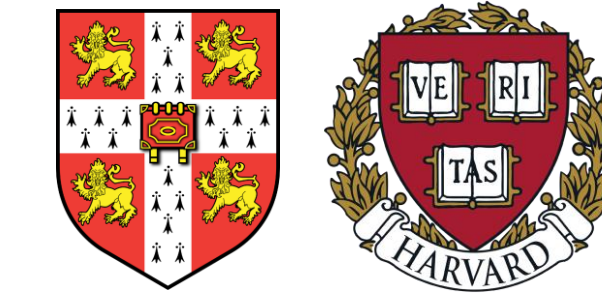


FairSSL: Fair Multimodal Self-Supervised Learning

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Project page



icml26-fairssl.github.io

1. Motivation

- Multimodal SSL often assumes that different views share enough task-relevant information.
- However, in heterogeneous real-world data, modalities can have **different lengths** and **temporal structure**.
- Strict cross-modal alignment can discard modality-specific signals and weaken fairness behavior.

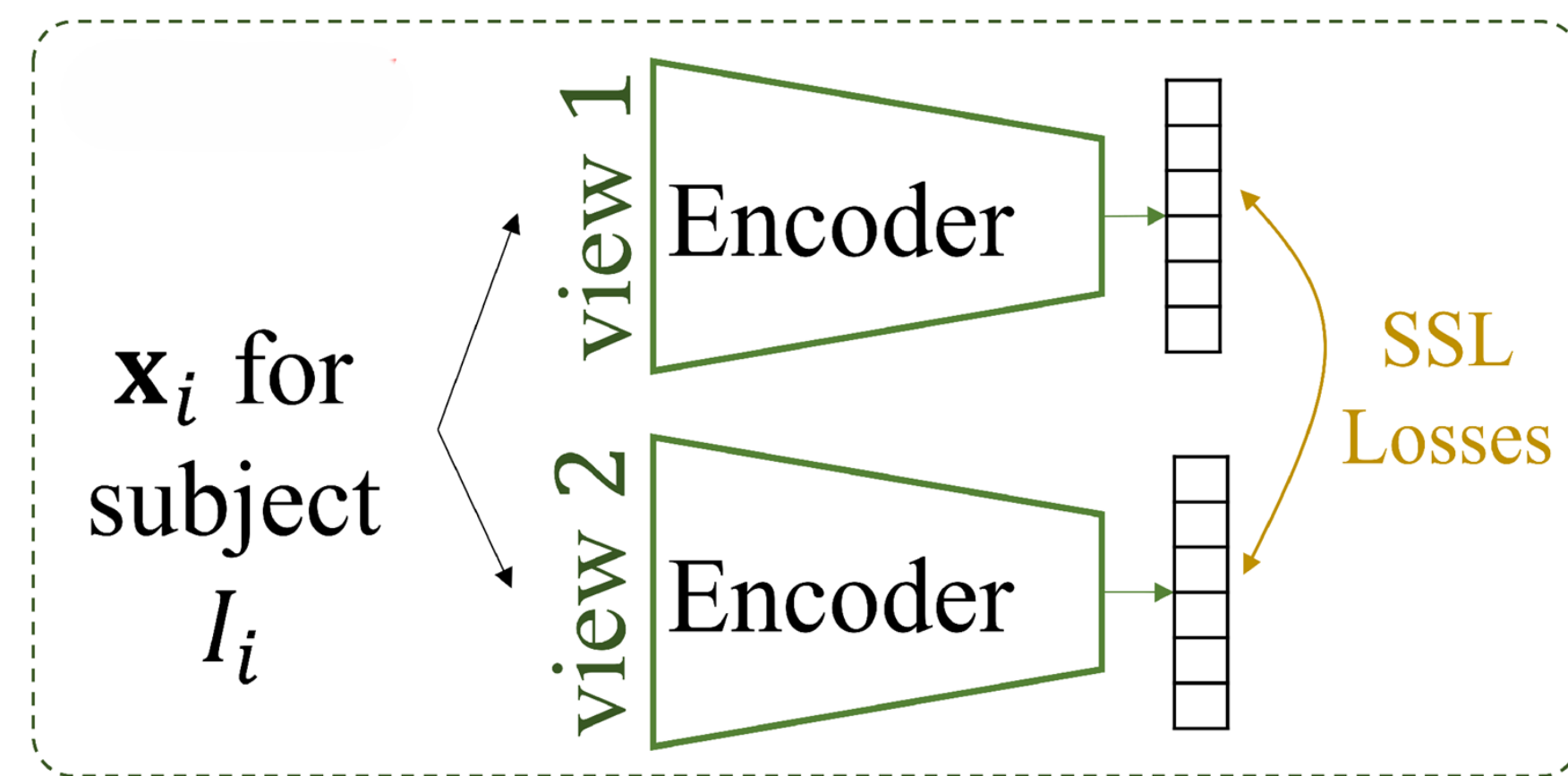


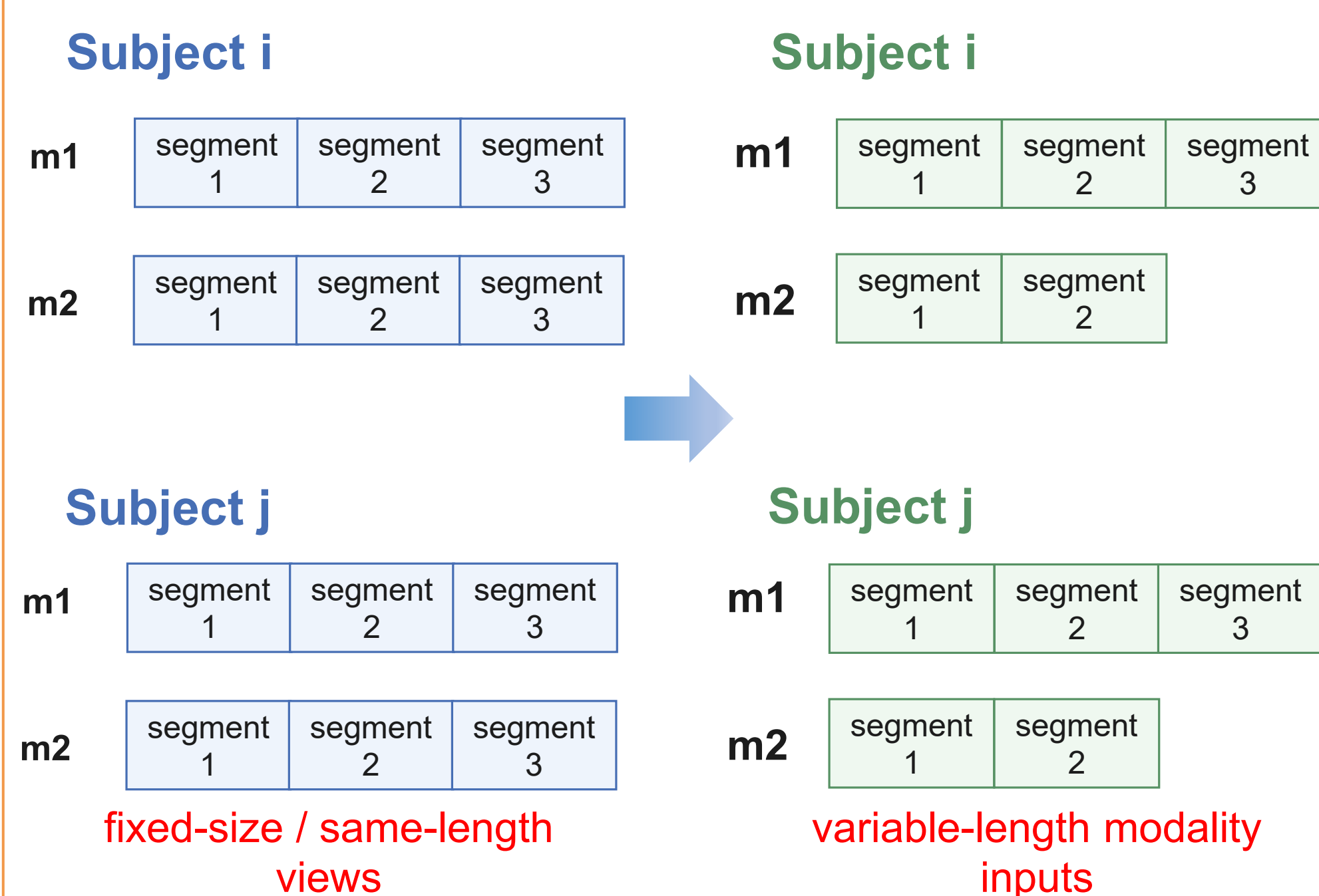
Figure 1: Existing work on Multimodal SSL^[2,5]

2. Variable-length multimodal subjects

- Each subject may have a different number of segments in each modality.
- One-to-one segment alignment between modalities is not always well-defined.
- FairSSL** treats this heterogeneity as useful structure rather than as noise.

Common SSL setup

FairSSL setting



m1, m2: different modalities

Figure 2: FairSSL handles subjects whose modalities contain different numbers of segments. ^[3,4]

3. VICReg^[1] Overview

- Variance:** prevents representation collapse by keeping embedding dimensions active:

$$V_{reg}(\{z_i\}) = \frac{1}{d} \sum_{j=1}^d \max\left(0, \gamma - \sqrt{\text{Var}(\{z_{i[j]}\})} + \varepsilon\right)$$

- Covariance:** reduces redundancy by decorrelating different embedding dimensions:

$$C_{reg}(\{z_i\}) = \frac{1}{d} \sum_{j \neq k} [\text{Cov}(\{z_i\})]_{j,k}^2$$

- Invariance:** makes representations of paired views similar:

$$I_{reg}(\{z'_i\}, \{z''_i\}) = \frac{1}{n} \sum_i \|z'_i - z''_i\|_2^2$$

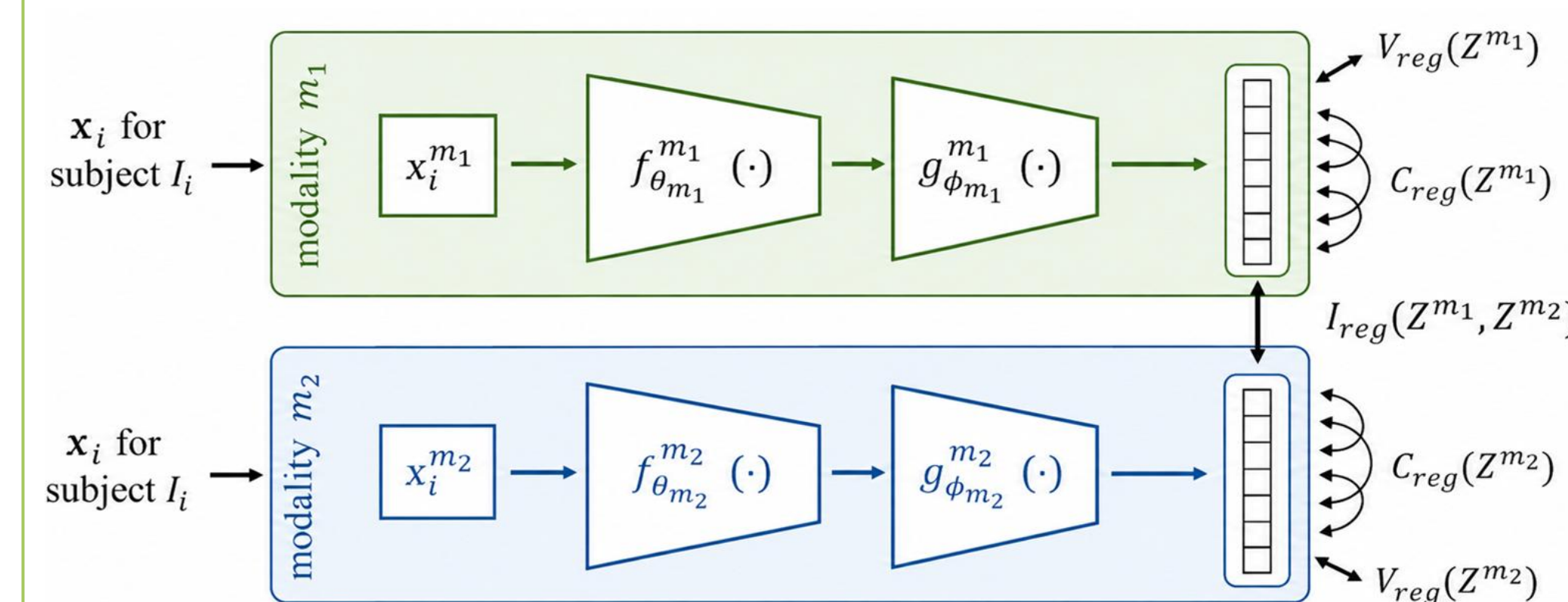


Figure 3: VICReg's joint embedding architecture with variance, covariance and invariance regularization terms.

4. FairSSL overview

- Encode each segment using a modality-specific encoder.
- Pool one modality into a subject-level representation.
- Align the pooled representation with available segments from the other modality.

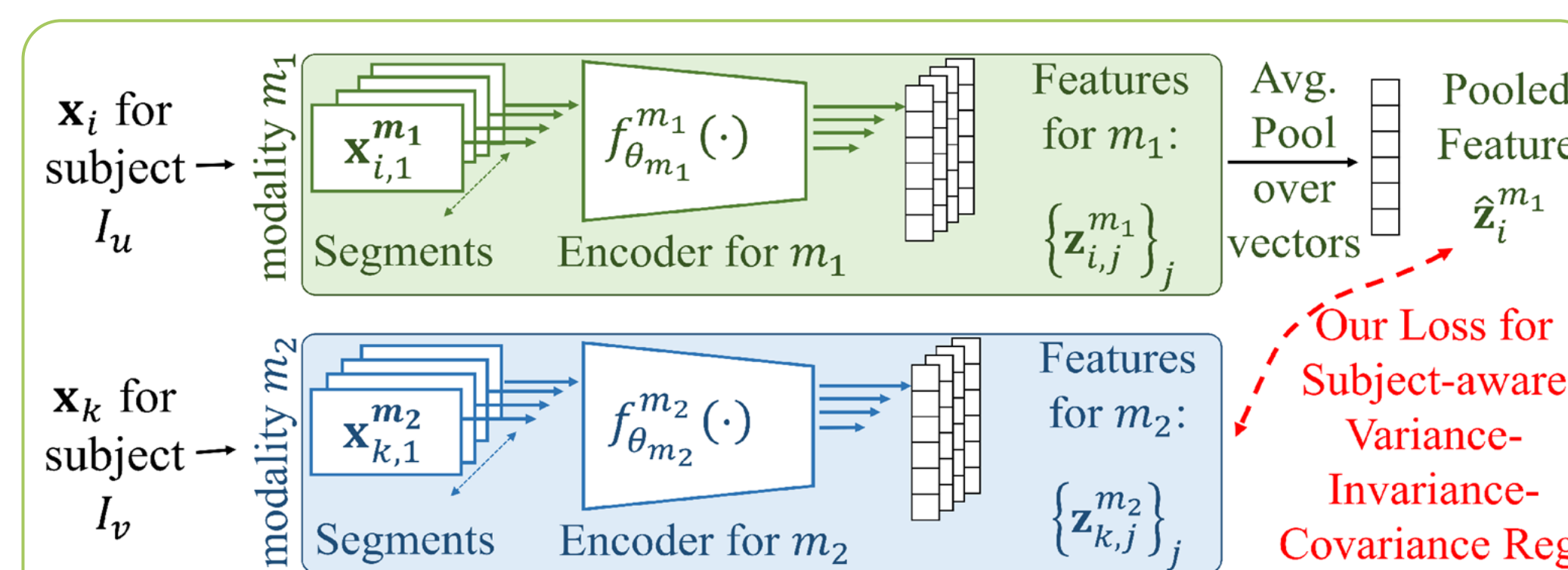


Figure 4: FairSSL processes each modality for the same or different subjects.

5. FairSSL's invariance variants

The variants differ only in which second-modality segments are aligned with the pooled representation.

M1: Intra-subject

$$I_{M1}^{FSSL} = \frac{1}{|B|} \sum_{i \in B} \lambda I_{reg}^{FSSL}(\hat{z}_i^{m_1}, z_i^{m_2})$$

Aligning subject i with subject i .

M2: Inter-subject

$$I_{M2}^{FSSL} = \frac{1}{|B|^2} \sum_{i, l_k \in B} \lambda I_{reg}^{FSSL}(\hat{z}_i^{m_1}, z_k^{m_2})$$

Align subject i with all subjects in the batch.

M3: Class-based

$$I_{M3}^{FSSL} = \frac{1}{|B|^2} \sum_{i, l_k \in B} \lambda I_{reg}^{FSSL}(\hat{z}_i^{m_1}, z_k^{m_2}), y_i = y_k$$

M2 alignment for same-class subjects.

M4: Alternating

$$I_{M4}^{FSSL} = \begin{cases} L_{M2}^{FSSL}, & e \bmod 2 = 1 \\ L_{M3}^{FSSL}, & e \bmod 2 = 0 \end{cases}$$

Alternate between M2 and M3 per epoch.

6. Why FairSSL improves fairness

Variance and **covariance** terms keep the within-group scores from collapsing by introducing a lower bound.

$$\text{Within-Group Score Spread} \leq O(I_{reg}^{FSSL})$$

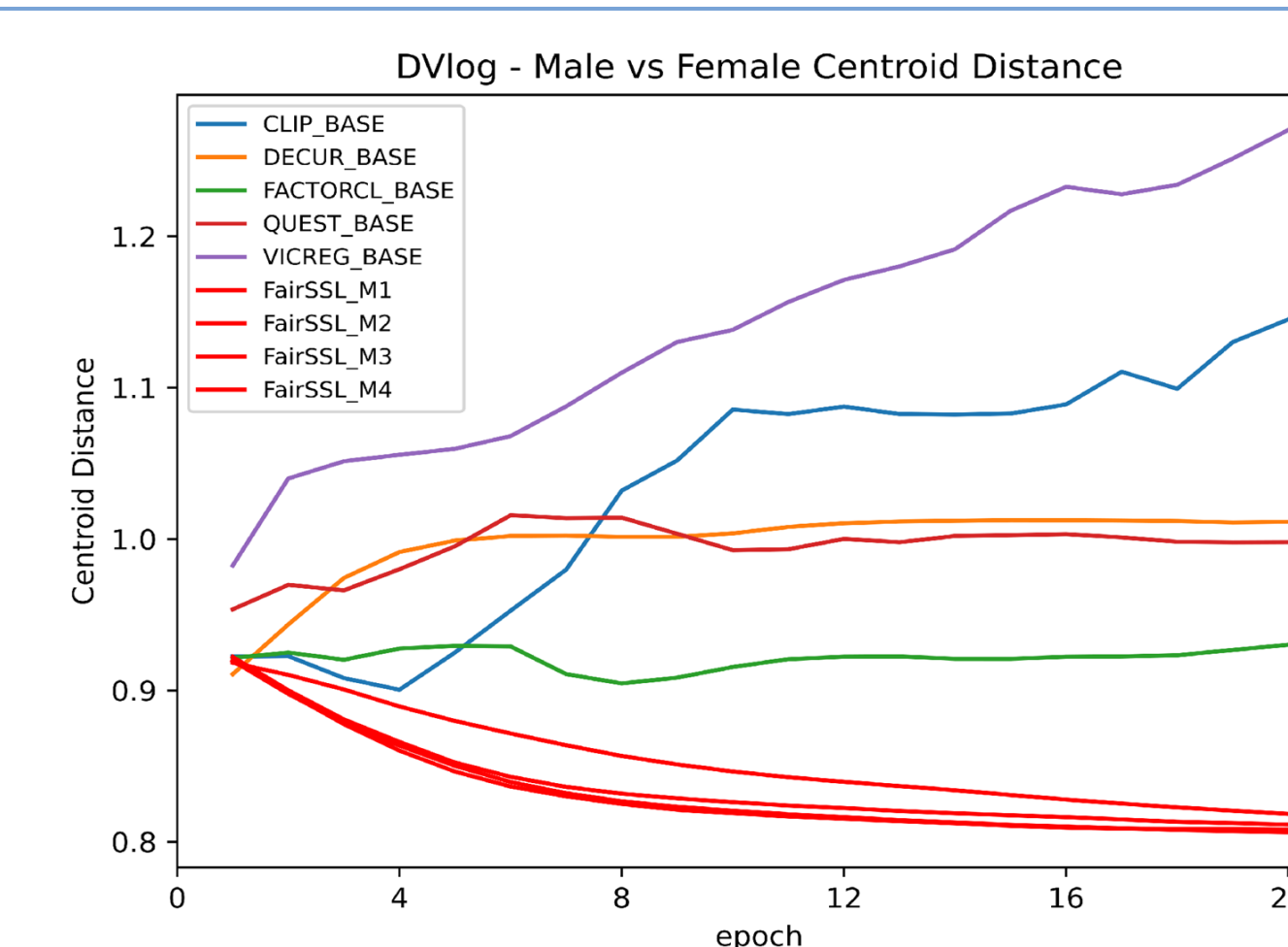
Invariance term reduces the between-group score disparity and upper-bounds within-group score deviation.

$$\Omega(f_{V_{reg}, C_{reg}}) \leq \text{Within-Group Score Spread}$$

Between-group score disparity and within-group score spread together bound how far group fairness metrics are from ideal.

$$|R_{metric} - 1| \leq O(\text{Within-Group Score Spread}) + O(\text{Between-Group Score Disparity})$$

Figure 5: Euclidean distance between the two gender centroids in a 2D PCA projection of the embeddings.



7. Main Results

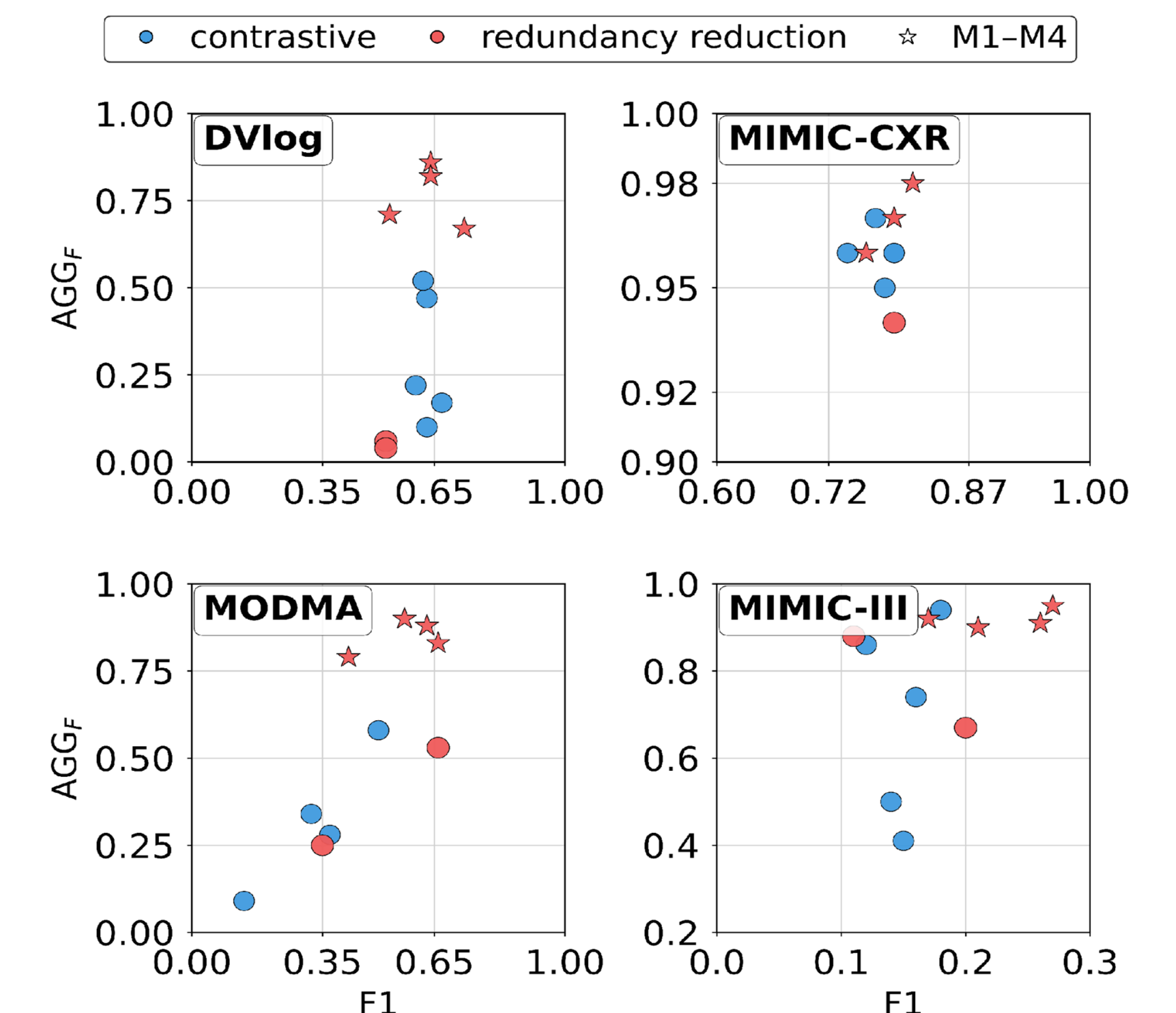
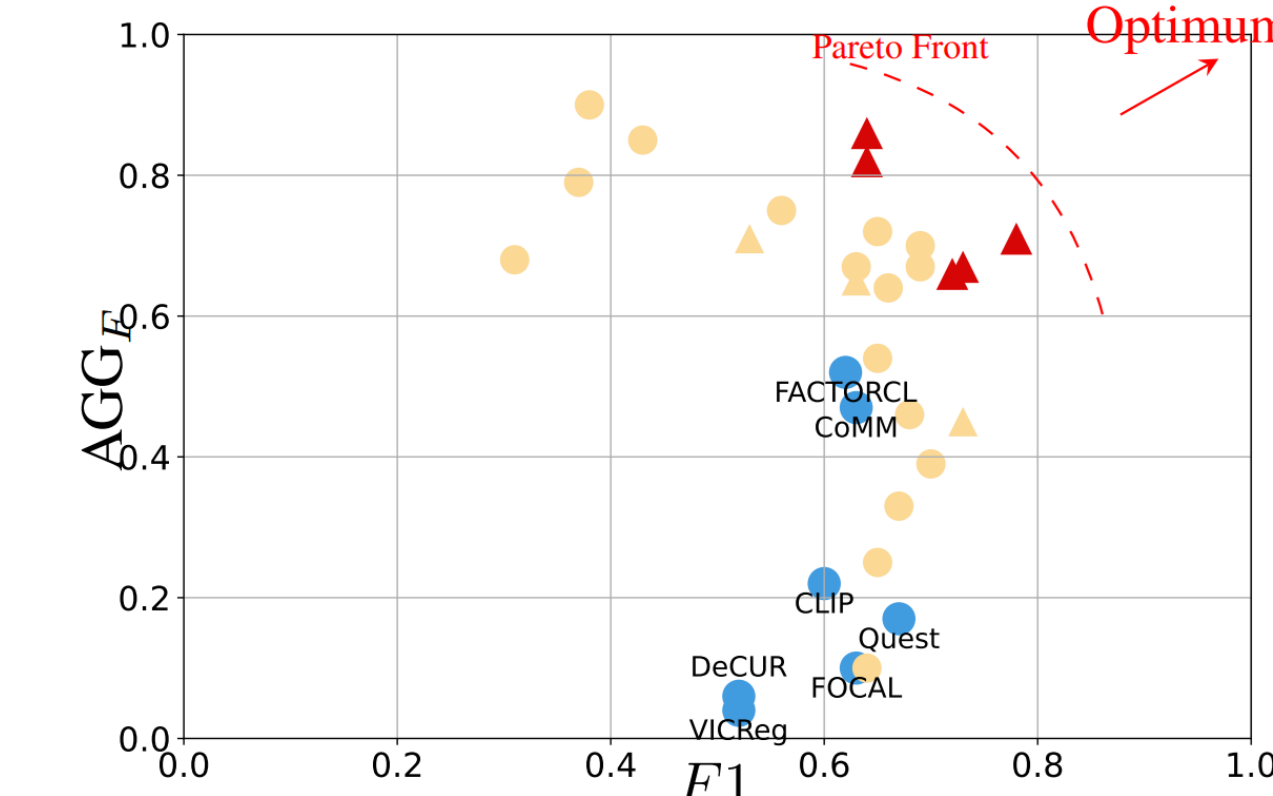


Figure 6: Main results on 4 datasets.

Figure 7: Pareto Plot for DVlog. Red triangles represent FairSSL. Blue circles represent baseline SSL. Yellow circles represent baselines with FairSSL.



- Supervised and SSL baselines can perform well, but often show weaker fairness behavior.
- FairSSL variants move closer to better fairness-performance trade-offs.

8. Conclusion

- FairSSL adapts SSL to heterogeneous, variable-length multimodal data.
- Segment pooling makes cross-modal alignment well-defined.
- Subject-aware invariance improves fairness without requiring strict segment matching.

References

- [1] Bardes et al., VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning, ICLR 2022.
- [2] Chen et al., "A Simple Framework for Contrastive Learning of Visual Representations," ICML 2020.
- [3] Liang et al., "Factorized Contrastive Learning: Going Beyond Multi-view Redundancy," NeurIPS 2023.
- [4] Liu et al., "FOCAL: Contrastive Learning for Multimodal Time-Series Sensing Signals in Factorized Orthogonal Latent Space," NeurIPS 2023.
- [5] Radford et al., "Learning Transferable Visual Models from Natural Language Supervision," ICML 2021.